

Project Title: Olist Product Sales on E-Commerce Analysis

Disclaimer: This is a personal project created solely for educational and portfolio purposes. The data used is a public dataset obtained from [Kaggle](#) and reflects only a specific time period. This project is not affiliated with, sponsored by, or representative of the internal data or official views of any related companies.

Background

E-commerce platforms continuously generate massive amounts of transactional data on a daily basis. To stay competitive, it is crucial for these companies to deeply understand various aspects of their operations, such as how customers behave, how well sellers perform, which products drive sales, and how efficiently deliveries are handled. Mastering these elements is the key to simultaneously boosting revenue and enhancing overall customer satisfaction.

To address these business needs, this project leverages PostgreSQL to conduct an in-depth analysis of the Olist Brazilian E-Commerce Dataset. By transforming this raw transactional data into actionable business insights, the analysis aims to provide a solid foundation for strategic, data-driven decision-making within the e-commerce space.

Project Selection Criteria

Before starting the analysis process, several criteria were established to ensure that the project would provide meaningful insights, maintain data credibility, and simulate real-world data analytics practices. The project selection was based on the following considerations:

1. The dataset must be publicly accessible and not contain private or confidential information.
2. The analysis object should represent an industry or topic that is widely recognized and easy to understand by general audiences.
3. The data source must come from trusted and credible platforms to ensure that the validity and reliability of the secondary data can be justified.
4. The project documentation and workflow will follow the Google Data Analytics framework, consisting of Ask, Prepare, Process, Analyze, Share, and Act.

Business Problem

1. Which products generate the highest sales?
2. Which sellers contribute the most revenue?
3. How satisfied are customers?
4. How efficient is the delivery process?
5. How can customer retention be improved?

Objective

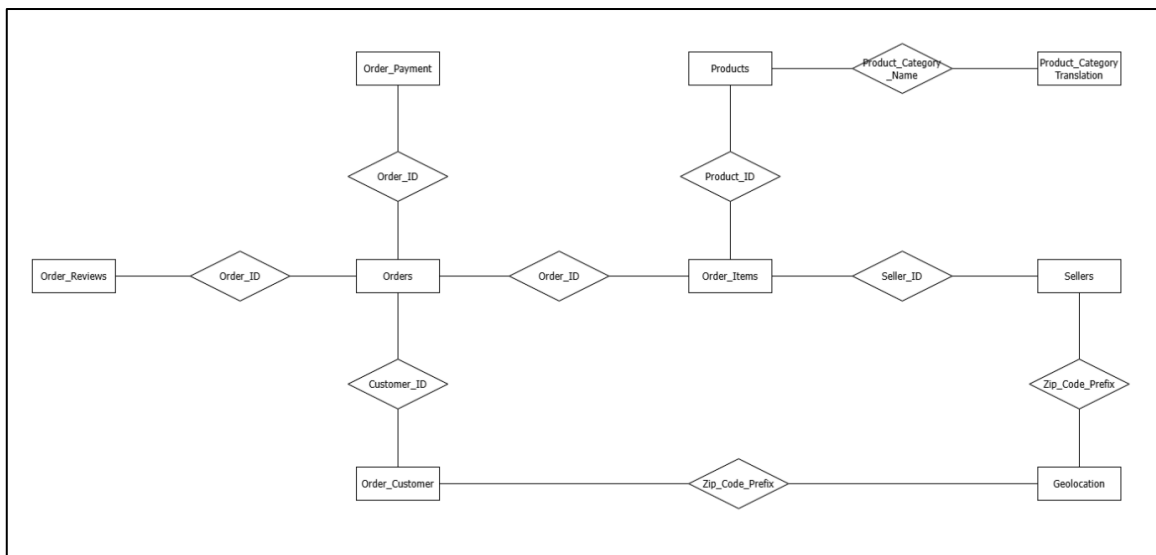
- Analyze sales performance
- Evaluate seller contribution
- Assess customer satisfaction
- Measure delivery efficiency
- Generate actionable recommendations

Dataset

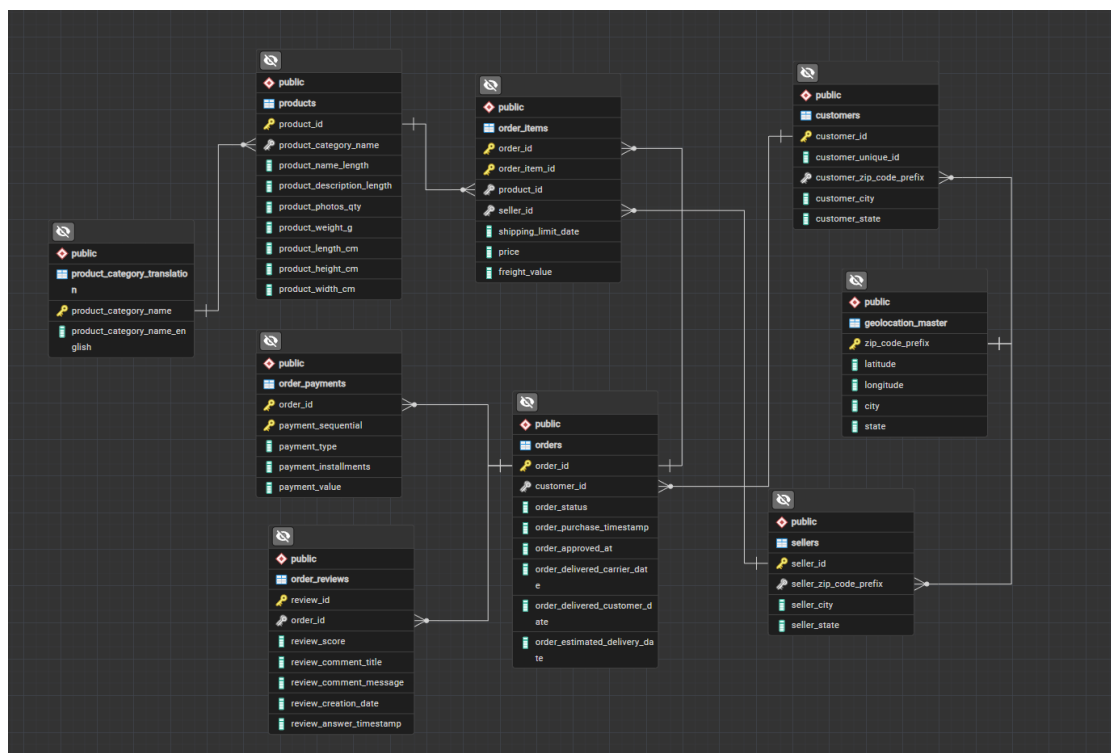
- Source:
Olist Brazilian E-Commerce Dataset
- Records:
 - Orders: 99,441
 - Customers: 99,441
 - Products: 32,951
 - Sellers: 3,095

Data Model

Entity Relationship Diagram (ERD)



Data Schema



The database consists of:

- Customers
- Orders
- Order Items
- Products
- Sellers
- Payments
- Reviews
- Geolocation
- Product Category Translation

Data Quality Assessment

Missing Values

Column	Missing
Product Category	610

Duplicate Records

No duplicate primary keys were found.

Referential Integrity

All foreign key relationships were validated successfully.

SQL Techniques Used

SQL Techniques:

- JOIN
- CTE
- Window Functions
- CASE WHEN
- Aggregations
- Subqueries
- Data Cleaning
- Referential Integrity Validation

Methodology

This project follows the Google Data Analytics framework. Each stage was documented to simulate a real-world data analytics workflow, ensuring a structured and comprehensive approach to solving the business problem. The methodology consists of six key phases:

1. **Ask:** Defining the primary business problem, understanding stakeholder objectives, and formulating the key questions regarding app performance and user satisfaction.
2. **Prepare:** Gathering the dataset from Kaggle, understanding its structure, and assessing data integrity to ensure it is relevant and sufficient for the analysis.

3. **Process:** Cleaning and transforming the raw data using Python and Pandas. This includes handling missing values, standardizing text, and preparing the review content for categorization.
4. **Analyze:** Examining the processed data to identify patterns and trends. In this phase, user reviews are categorized into the seven predefined business areas, and rating distributions and temporal trends are calculated.
5. **Share:** Creating compelling data visualizations and building an interactive dashboard in Google Looker Studio to effectively communicate the insights to stakeholders.
6. **Act:** Formulating actionable, data-driven recommendations based on the findings to help Shopee management prioritize service improvements and enhance user experience.

Analysis Process

Analysis Process:

1. Database Design
2. Data Import
3. Data Profiling
4. Data Cleaning
5. Exploratory Data Analysis
6. Business Analysis
7. Insight Generation
8. Recommendations

Key Findings

After analyzing the processed dataset, several significant patterns and insights emerged:

- Data quality analysis revealed 10 geolocation records located outside of Brazil, as well as 610 products labeled as 'Uncategorized.' To preserve audit trails and referential integrity, no out-of-bounds records were physically deleted from the system.
- Based on the Total_Revenue data, the platform generated a precise total of 16.008.872,12 BRL or equal to 31.766.097,36 USD. Coupled with an Average Order Value of 160.99 BRL or equal to 31,96 USD, this indicates a very healthy transaction volume and a strong market presence.
- The total number of sellers on the platform is 3,095. However, calculating the sum of the revenue from just the top 10 sellers (Top10_Seller) yields approximately 1.787.241,74 BRL or equal to 354815,92 USD. Dividing this by the total revenue (16 million BRL) reveals that a tiny fraction of the seller base (0.3%) is responsible for over 11% of the entire platform's income.
- The Total_Data_Each_Table shows there are 99,441 unique customers in the database. The Total_User_per_Segment query reveals that exactly 99,441 users fall into the "One-Time" buyer segment. This means 100% of the customer base in this dataset only made a single purchase, exposing a severe issue with customer retention.
- The Customer_Satisfaction query resulted in an average score of 4.088 out of 5. While a score above 4 is generally positive and indicates decent product quality, it suggests there are still notable friction points in the customer journey preventing higher satisfaction levels.

- The `Average_Delivery_Performance` query highlights that the average time from order placement to delivery is over 12.5 days (12 days 13:24:31). In the e-commerce industry, a nearly two-week wait time is a major pain point and is highly likely the primary factor capping the average review score at ~4.09.

Recommendation

Based on the key findings, the following actionable recommendations are proposed for Shopee's management and technical teams:

1. With 100% of the recorded user base being one-time buyers, acquiring new customers is currently the only source of revenue. The business must immediately implement retention strategies such as loyalty points, personalized email marketing, and post-purchase discount vouchers to encourage repeat purchases and improve Customer Lifetime Value (CLV).
2. An average delivery time of over 12 days is a significant operational bottleneck. To elevate the average review score above 4.5, management should evaluate current logistics partners, consider adding strategic fulfillment centers closer to high-density customer areas, or provide premium expedited shipping options.
3. Since just 10 sellers (out of over 3,000) are driving more than 11% of the total revenue, these merchants are critical to the platform's success. Olist should offer these VIP sellers dedicated account managers, reduced commission tiers, or premium placement on the platform to ensure their continued partnership and stock availability.
4. By leveraging the `geolocation_master` table alongside revenue data, the marketing team should identify the specific cities and states that yield the highest Average Order Value and direct their advertising budget toward acquiring new users in those specific, highly profitable areas.

Conclusion

The analysis identified significant opportunities to improve customer retention and operational efficiency. While revenue performance was strong, customer loyalty remains relatively low, indicating potential for future growth through retention-focused initiatives.

Future Improvements

- Implement RFM Customer Segmentation.
- Conduct Cohort Analysis.
- Develop a PostgreSQL Data Warehouse.
- Create automated reporting pipelines.